Answer key: Lesson 1, Bundles and Singles

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Count the sticks.

Answer: 2 bundles, 6 singles, 26 sticks

Count the 2 bundles in tens: 10, 20. Then count on the 6 singles: 21, 22, 23, 24, 25, 26. So 2 bundles, 6 singles, 26 sticks.

Watch for: Counting each bundle as 1 (2 + 6 = 8).; Writing the digits the wrong way round (62).

S2. Draw 3 bundles of ten and 5 single sticks in the box. Then write the number.

Answer: 35 (3 bundles drawn, each with ten sticks, and 5 singles)

Draw 3 rings each holding ten short lines (that is 30), then 5 single lines. Count: 10, 20, 30, 31, 32, 33, 34, 35.

Watch for: Drawing 3 sticks and 5 sticks instead of 3 bundles of ten.; Writing 53.

S3. Write the number.

Answer: 42, 70, 19, 85

(a) 4 tens is 40, and 2 more is 42. (b) 7 tens is 70 and there are no ones: 70. (c) 10 and 9 is 19. (d) 8 tens is 80, and 5 ones makes 85.

Watch for: Writing 7 for 7 tens and 0 ones.; (d) writing 58 because 5 was said first.

S4. Fill in the tens and ones.

Answer: 5 tens 8 ones; 4 tens 0 ones; 7 tens 1 one

(a) 58: the 5 is the tens digit, the 8 is the ones digit. (b) 40: 4 tens and no ones. (c) 71: 7 tens and 1 one.

Watch for: Saying 40 has 4 ones.; Swapping the two digits.

S5. Ali has 47 straws. He ties them into bundles of ten.

Answer: 4 bundles, 7 left over

47 is 4 tens and 7 ones, so Ali makes 4 full bundles and has 7 straws left over (4 bundles hold 40 straws).

Watch for: Answering 47 for the bundles.; Saying 3 bundles and 17 left over: 17 singles can still make another bundle.

S6. Siti has 3 bundles of ten and 4 single sticks. She writes 43. Is Siti right? Write the number she should write, and say what she mixed up.

Answer: No. 34. She wrote the tens and ones digits the wrong way round (43 is 4 tens and 3 ones).

3 bundles is 30 and 4 singles makes 34. Siti wrote 43, which is 4 tens and 3 ones. The tens digit goes on the left.

Watch for: Saying she is right because the digits 3 and 4 are there.; Not saying what she mixed up (any clear sentence about the digits is fine).

S7. Ali has 6 bundles of ten and no single sticks. Mei has 6 single sticks and no bundles. Draw what each child has in the box. Then write each number.

Answer: Ali 60, Mei 6. Ali has more (6 tens is 60; 6 singles is 6)

Ali: 6 bundles of ten is 60, with no singles, so 60. Mei: 6 single sticks is 6. The same digit 6 means very different amounts because of its place: 6 tens or 6 ones.

Watch for: Writing 6 for both children.; Drawing 6 sticks for Ali instead of 6 bundles of ten.

Think harder (page 3)

T1. Mei has 5 bundles of ten and 14 loose sticks. How many sticks has she? Hint: can she make another bundle?

Answer: 64 (14 loose sticks make 1 more bundle and 4 left, so 6 tens and 4 ones)

14 loose sticks = 1 bundle and 4 singles. So Mei has 5 + 1 = 6 bundles and 4 singles: 64 sticks. Check: 50 + 14 = 64.

Watch for: Writing 514 by joining the digits.; Writing 5 tens and 14 ones as 5 + 14 = 19.

T2. 41 is 4 tens and 1 one. Open one bundle. Now 41 is 3 tens and ___ ones.

Answer: 11 (3 tens is 30, and 30 + 11 = 41)

Opening a bundle gives 10 more singles. 1 single + 10 singles = 11 singles, and 3 bundles are left. 30 + 11 = 41, so it is the same number.

Watch for: Writing 10 (forgetting the 1 single already there).; Writing 1 (forgetting the opened bundle).

T3. Mystery number. Clue 1: I am between 30 and 50. Clue 2: my ones digit is 6. Clue 3: my number of tens is even. What number am I?

Answer: 46

Between 30 and 50 and ending in 6: 36 and 46. 36 has 3 tens (odd). 46 has 4 tens (even). So the number is 46.

Watch for: Stopping at 36 and not checking clue 3.; Listing numbers that start with 6.

T4. Mei has 7 tens and 2 ones. Ali has 6 tens and 12 ones. Who has more, or do they have the same? Show how you know.

Answer: The same: 72 each (6 tens 12 ones is 7 tens 2 ones)

Ali's 12 ones make 1 more bundle and 2 ones. So Ali has 7 tens and 2 ones, the same as Mei. Both have 72.

Watch for: Saying Ali has more because 12 is bigger than 2.; Adding 6 + 12 = 18 and comparing 18 with 9.

Go further (page 4)

G1. Ben has 6 bundles of ten and 14 loose sticks. Siti has 85 loose sticks. Who has more sticks? How many more?

Answer: Siti. Ben has 74, Siti has 85, so Siti has 11 more.

Ben: 6 tens and 14 ones is 7 tens and 4 ones, so 74. Siti: 85. 85 is 8 tens 5 ones and 74 is 7 tens 4 ones: 1 ten and 1 one more, which is 11 more.

Watch for: Comparing only 6 and 85 without working out Ben's whole number.; Not opening 14 ones into a bundle before comparing.

G2. A shop sells pencils in boxes of ten. Mei needs 60 pencils. She has 4 full boxes and 18 loose pencils. How many more pencils does she need?

Answer: 2 more pencils (4 boxes and 18 pencils is 58)

4 boxes is 40 pencils. 40 + 18 = 58. She needs 60, so she needs 60 - 58 = 2 more.

Watch for: Saying 18 more (the loose pencils).; Forgetting the 4 full boxes hold 40 pencils.

G3. How many different ways can you show 30 with bundles of ten and loose sticks, if you may open bundles? Write each way as ___ tens and ___ ones. (3 tens and 0 ones is one way.)

Answer: 4 ways: 3 tens 0 ones; 2 tens 10 ones; 1 ten 20 ones; 0 tens 30 ones

3 tens 0 ones. Open one bundle: 2 tens 10 ones. Open another: 1 ten 20 ones. Open the last: 0 tens 30 ones. That is 4 ways. Each time one ten becomes 10 ones.

Watch for: Stopping at 3 tens 0 ones and 0 tens 30 ones (2 ways).; Counting 2 tens 10 ones twice.

Investigation (page 5): Swap the digits

Three things can happen. If the tens digit is smaller than the ones digit (14, 26), swapping makes a bigger number (41, 62). If the tens digit is bigger (52, 91), swapping makes a smaller number (25, 19). If the two digits are the same (77), the number stays the same.

Look for: A child who tests several numbers, says what they notice and gives a reason using tens and ones. A number that ends in 0 (30) becomes 03, which is just 3, so it is smaller: this fits 'tens digit bigger'.

Exit ticket (page 5)

E1. How many sticks?

Answer: 48

4 bundles is 40, and 8 singles makes 48.

Watch for: Writing 84.; Writing 12 (4 + 8).

E2. Fill in the tens and ones.

Answer: 3 tens and 7 ones

The left digit 3 counts the tens, the right digit 7 counts the ones.

Watch for: Swapping: 7 tens and 3 ones.

E3. Write the number: 7 tens and 0 ones. Why do we need the 0?

Answer: 70. The 0 shows there are no ones, and it keeps the 7 in the tens place (without it, it would look like 7 ones).

7 bundles of ten and no singles is 70. The zero holds the ones place.

Watch for: Writing 7.; Writing 700 or 7 0 as 7 + 0.

Sneak-peek puzzles (not in the workbook)

1. 24 stars.
2. Yes. 3 bundles of ten is 30, and 4 more is 34.